

2nd FTC

Section 4.4

11/3/25

$$\frac{d}{dx} \left[\int_1^x t^2 dt \right] = \frac{d}{dx} \left(\frac{1}{3} t^3 \right) \Big|_1^x = \frac{d}{dx} \left(\frac{1}{3} x^3 - \frac{1}{3} \right) = x^2$$

$$\frac{d}{dx} \left(\int_{\pi/6}^x \cos t dt \right) = \frac{d}{dx} (\sin t) = \frac{d}{dx} \left(\sin x - \frac{1}{2} \right) = \cos x$$

2nd FTC

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

$$\frac{d}{dx} \int_x^a f(t) dt = -f(x)$$

$$\begin{aligned} \frac{d}{dx} \int_{\pi/6}^{x^2} \cos t dt &= \frac{d}{dx} (\sin t) \Big|_{\pi/6}^{x^2} = \\ &= \frac{d}{dx} \left(\sin x^2 - \frac{1}{2} \right) = \\ &= 2x \cos(x^2) \end{aligned}$$

Chain 2nd FTC

$$\frac{d}{dx} \left(\int_a^{g(x)} f(t) dt \right) = f(g(x)) \cdot g'(x)$$

$$\frac{d}{dx} \int_3^x \sqrt{1+t^2} dt = \sqrt{1+x^2}$$

$$\frac{d}{dx} \int_2^x \tan(t^3) dt = \tan x^3$$

$$\frac{d}{dx} \int_{-1}^{x^3} \frac{1}{1+t} dt = \frac{(3x^2)}{1+x^3}$$

$$\frac{d}{dx} \int_2^{\sin x} \sqrt[3]{1+t^2} dt =$$

$$3\sqrt[3]{1+\sin^2 x} \cdot (\cos x)$$